X-PIXEL

Next-Generation Cross-Platform Optical Communication Infrastructure

Abstract

This document serves as the research foundation for X-Pixel — a next-generation cross-platform optical communication infrastructure enabling ultra-secure, fully offline file transfer between Android and iOS devices using visual optical transmission alone.

1. Introduction

X-Pixel transforms smartphone screens into optical transmitters and cameras into intelligent optical receivers. The system leverages Screen-Camera Communication (SCC), Optical Camera Communication (OCC), rolling shutter exploitation, spectral-domain embedding, and neural-network-assisted decoding.

2. Multi-Layer Encoding Strategy

Layer	Technique	Throughput	Use Case
Layer 0	Dynamic QR Stream	~320 kbps	Compatibility
Layer 1	RGB Color DFC + DCT	~225 kbps	Standard transfer
Layer 2	Blue-Channel Modulation	~1.2 kbps	Imperceptible mode
Layer 3	Rolling Shutter OCC	~50-200 kbps	High-speed transfer

3. Novel Contributions

Key research contributions include:
- Cross-platform Android ↔ iOS optical protocol
- Adaptive multi-modal optical communication
- Hybrid Rolling Shutter + DFC modulation
- Multi-region optical MIMO encoding
- Production-grade mobile OCC SDK

4. Future Directions

Future research directions include optical mesh networking, transformer-based neural decoders, GAN-assisted modulation optimization, and near-infrared invisible communication channels.

End of Document — X-Pixel Research Foundation